Discrete Models in Biology Homework
Modelling invasive Glioma Cells with a Cellular Potts Model

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1 Introduction

Glioblastoma is a type of cancer originating from glial cells in the brain that is very invasive. In their 2016 article, A. Fayzullin *et al.*, used time-lapse microscopy technics to study the behavior of invasive glioma cells originating from a tumor grafted on rodent brain slice. They classified the gliomal cell in two class of phenotype. The non-invasive cells stayed in the tumorsphere, while invasive cells migrated away from the bulk of the tumor. They observed very specific migratory movements, with some types of invasive cells displaying straight radial trajectories away from the core, even after multiple day of invasion, and up to a millimeter of distance between the invasive cells and the core. Upon complete removal of the tumor core, the invasive cells stopped their centrifugal trajectories and displayed random movement. However, after incomplete removal of the core, some invasive cell reversed their direction and migrated back to the tumor and participated in the reconstruction of the core.

As hypothesized in the article, these two experiments suggest some sort of signalling between the tumorsphere and the migrating invasive cells. In this project, I first present a possible mechanism that could explain the behavior observed in [1].

To test if this mechanism is sufficient to explain empirical data, an *in silico* model must be built. I choose to start from a Cellular Potts model for three reasons. First, they are quite general, have a natural way to represent cell/cell adhesion surface, and are known to couple well with diffusion-reaction PDE models. Second, they are pedagogically interersting as I've never used them, contrary to agent-based models. Finally, there are records in the scientific literature of previous use of this type of model to study the behavior of glioblastoma.

In their article from 2008, B. M. Rubenstein and L. J. Kaufman, tested the impact of the extracellular matrix on the tumor using a cellular potts model with a hamiltonian consisting of a term for adhesion potentials and an elastic potential driving the cell toward a target perimeter. Their cells were also able to divide upon reach of their target perimeter. Overall, this model focused on the evolution of the tumor core and didn't account for the invasive cell phenotype.

In 2013, A. Szabó and R. M. H. Merks used a cellular potts model to study invasive tumor. This time they consider adhesion potentials and an elastic potential driving the area of the cells toward a target area. They extended the model to account for chemotaxis. The diffusion of chemical signals is computed by solving a diffusion equation, and, at each step of the Metropolis-Hasting algorithm, the difference in potential energy is biased proportionally to the difference of signal concentration in the source and destination points.

I then present in details the model that was implemented, which was simplified because of time constraints.

2 A possible mechanism

Here I describe in biological term, a possible mechanism susceptible to explain the behavior observed by A. Fayzullin *et al.*.

Glioma cells can express two different phenotypes, invasive (I) and non-invasive (N), and communicate using two chemical signals that are diffused in the medium: the stress (S) signal, and the attractive (A) signal. In low stress environment, N cells divide and emit signal A. Signal S is always emitted by N cells. In high stress environment, N cells stop emitting signal A. In presence of signal A, cells I climb up the concentration gradient of A. In absence of this signal, cells I move down the concentration gradient of signal S. I cell change their phenotype to N in low stress and high density environment, N cells change their phenotype to I in high stress and low density setting. This hypothesis is summarized in Figure 1.

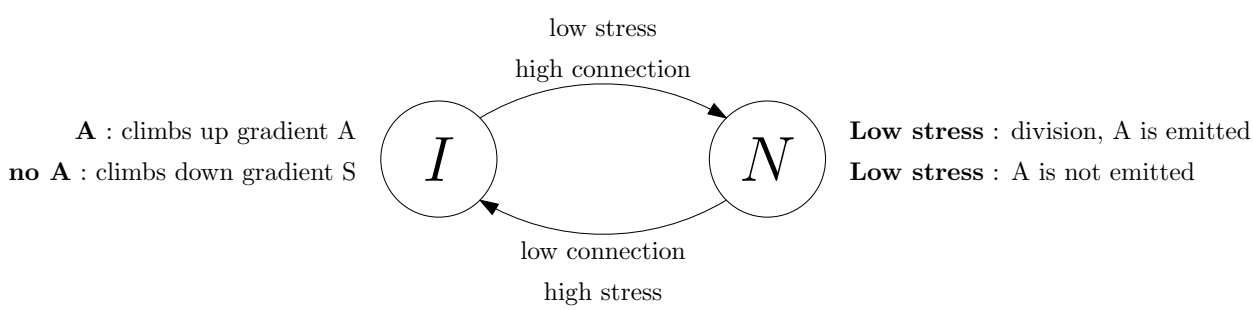

Figure 1: Diagram of the proposed biological mechanism

This model could explain the behavior observed empirically. Upon grafting of a tumor core composed of tightly packed N cells, S signal would rapidly build up and form a radial concentration gradient around the tumor. Cells at the periphery of the tumor, being less connected, would start expressing phenotype I and would start migrating radially from the tumor. Removal of the tumorsphere would remove all significant sources of S or A signal, the absence of chemical gradient would result on random movement of remaining I cells. Removal of a part of the tumorsphere would decrease the equilibrium concentration of S in the tumor. Remaining N cells would start emitting signal A, which will cause nearby I cells to switch direction, go back to the tumor and switch to phenotype N once arrived because of the low S signal and increased cell density.

Note that, if we assume that cells are not able to produce both a positive and a negative gradient for the same chemical signal, we can consider that a single chemical signal can carry 1 bit of information. Three different configurations were shown to have different long range effect (1 : intact core; 2 : completely removed core; 3 : partially removed core), which corresponds to $\log_2(3)$ bits of information, which would require at least two chemical signals, under our previous assumption. This shows that, while the proposed model is complex, it may satisfy the principle of parcimony.

3 Model

The model takes place on a $W \times H$ lattice, and we denote by V the Von-Neumann neighborhood on this lattice, i.e. $(i, j) \in V$ if i and j differ by 1 on only one of their coordinates.

3.1 Continuous time Markov chain

The transition between the I and N phenotypes is modelled by a continuous time Markov chain (CTMC), modified to have parameters varying with the environment.

$$\mathbb{P}(\tau(c)_{t+dt} = N \mid \tau(c)_t = I) = t_{I \to N} l(c) dt \tag{3.1}$$

$$\mathbb{P}(\tau(c)_{t+dt} = I \mid \tau(c)_t = N) = t_{N\to I}(1 - l(c))dt \tag{3.2}$$

where $l(c)$ is the proportion of the cell's perimeter that is contacting another cell, and $t_{i\to j}$ is the base transition rate of cell type i to cell type j.

3.2 Hamiltonian

In the following we note $\sigma(i)$ the index of the cell in position i and $\tau(i)$ it's cell phenotype (I or N). We note C the set of all cells.

The Metropolis Hasting (MH) algorithm samples the system according to the Gibbs distribution :

$$g \propto e^{-\frac{H}{T}} \tag{3.3}$$

For this reason, we need to define the Hamiltonian of the system H. It's the sum of different energetical terms :

$$H = H_{\text{adh}} + H_{\text{area}} + H_{\text{perim}} + H_{\text{chemo}} \tag{3.4}$$

The adhesion term H_{adh} models the adhesion of cells and prevents them from disolving into each other.

$$H_{\text{adh}} = \sum_{(i,j)\in V} J_{\tau(i),\tau(j)} \mathbb{1}_{\sigma(i)\neq\sigma(j)} \tag{3.5}$$

where $J_{i,j}$ is the potential associated with one unit of surface contact between a cell of type i and a cell of type j.

We use an elastic potential H_{area} to drive the cells toward a target area.

$$H_{\text{area}} = \sum_{c\in C} \lambda_{\text{area}}(a(c) - a_t)^2 \tag{3.6}$$

where $a(c)$ is the area of cell c (i.e. number of lattice point taken by c), a_t is a target area, and λ_{area} is the elastic modulus for the area.

We use an elastic potential H_{perim} to drive the cells toward a target perimeter.

$$H_{\text{perim}} = \sum_{c\in C} \lambda_{\text{perim}}(p(c) - p_t)^2 \tag{3.7}$$

where $p(c)$ is the perimeter of cell c (i.e. number of point of c that have points not in c in their 4-neighborhood), p_t is a target perimeter, and λ_{perim} is the elastic modulus for the perimeter.

As in [3], chemotaxis is incorporated using the method from [4]. During a MH step, the change in potential energy is biased using the concentration s of chemical signal. If the state of point i is being copied into point j, we define the biased energy change as

$$\Delta H' = \Delta H + \xi_{\tau(i)}(s(j) - s(i)) \tag{3.8}$$

where $\xi_{\tau(i)}$ is the strengh of the chemotactic motion on cell type $\tau(i)$. $\xi > 0$ corresponds to a movement down the concentration gradient, and $\xi < 0$ up the concentration gradient.

Another way to incorporate chemotaxis that was experimented with is the following potential, which uses the everage concentration of the chemical signal inside the cell.

$$H_{\text{chemo}} = \sum_{c \in C} \mu_{\tau(c)} \frac{1}{a(c)} \sum_{i \in c} s(i) \tag{3.9}$$

where $\mu_{\tau(c)}$ is the strengh of the chemotactic motion on cell type $\tau(x)$. $\mu > 0$ corresponds to a movement down the concentration gradient, and $\mu < 0$ up the concentration gradient.

3.3 Diffusion equation

The concentration s of chemical signal S is controlled by a diffusion partial differential equation.

$$\partial_t s = D_s \nabla s - d_s s + e_s \mathbb{1}_N \tag{3.10}$$

Where D_s is the constant diffusion coefficient, d_s is the rate of decay, and e_s is the rate of emission. $\mathbb{1}_N$ is a function equal to 1 inside N cells and 0 everywhere else.

3.4 Simplified model

Because of time constraint, only a subset of this model could be implemented in time. The effect of signal A was not considred in the implementation of the model.

4 Implementation details

The model was implemented in C++, using the standard library, and raylib to build a graphical user interface.

The core of the cellular potts model is the following MH algorithm.

A Metropolis Hasting step

```
H is the value of the hamiltonian
(x, y) is sampled uniformly on the lattice
σ = L(x, y) is the state of (x, y) on the lattice L
σ' is sampled uniformly among the states of the neighbors of (x, y)
L(x, y) ← σ'
H' is the new value of the hamiltonian
r is sampled uniformly in [0, 1]
if r < exp(−(H' − H)/T)
    H ← H'
else
    L(x, y) ← σ
```

The main challenge of the implementation is to reduce the time required to compute the energy function at each iteration of the MH algorithm. To do this, the attributes of each cell (area, perimeter, etc.) are kept in memory. Then, during a transition of the system, only the cells affected by the change are updated, keeping the time complexity of lattice update constant.

The diffusion PDE is solved numerically on the same lattice as the Cellular-Potts model, using an explicit scheme.

$$s'_{x,y} = s_{x,y} + dt\Big(D_s\big(s_{x-1,y} + s_{x+1,y} + s_{x,y-1} + s_{x,y+1} - 4s_{x,y}\big) - d_s s_{x,y} + e_s \mathbb{1}_{\tau(x,y)=N}\Big) \tag{4.1}$$

At each time step of this scheme, the new value of s is written in a back buffer, which is then switched with the front buffer, which allows for a correct implementation of the numerical scheme with minimal data movement.

Updating the PDE is quite expensive compared to the rest of the implementation, with a time complexity in $\mathcal{O}(W \times H)$. It is not needed to update the PDE after each transition of the system. Therefore, the PDE is only updated with a specific frequency.

To initialize the tumor, a circular region of radius R of the lattice is set to be cells. Then, a basic k-mean algorithm is applied with $k =$`init_nb_cells`. This partitions the region according to a voronoi diagram in such a way that the sub-regions have similar size, each subregion is then initialized as a cell of type N.

To be able to test the system without cell phenotype transitions, another parameter `init_with_I` has been added. Any cell c that is initialized such that $l(c) <$ `init_with_I` is set as type I.

5 List of parameters

Table 1: List of model parameters

Parameter	variable name	role
$t_{I\to N}$	`t_ni`	base rate of $I \to N$ transition
$t_{N\to I}$	`t_in`	base rate of $N \to I$ transition
a_t	`target_area`	target area of the cells
λ_{area}	`lambda_area`	elastic modulus of the area force
p_t	`target_perim`	target perimeter of the cells
λ_{perim}	`lambda_perim`	elastic modulus of the perimeter force
$J_{I,I}$	`J_ii`	adhesion potential of one length unit between two I cells
$J_{N,N}$	`J_nn`	adhesion potential of one length unit between two N cells
$J_{I,N}$	`J_in`	adhesion potential of one length unit between a I and a N cell
D_s	`S_diffusion`	diffusion coefficient of signal S
d_s	`S_decay`	decay rate of signal S
e_s	`S_emit`	emission rate of signal S
ξ_I	`xi`	strengh of chemotactic motion of I cells
μ_I	`mu_S_inv`	strengh of chemotactic motion of I cells

Table 2: List of numerical parameters

Parameter	variable name	role
W	`lattice.width`	width of the lattice
H	`lattice.height`	height of the lattice
T	`T`	temperature of the sampling procedure
dt	`S_dt`	time discretization for the PDE solver
	`S_step_frequ`	frequency of update of the PDE
	`CTMC_dt`	time discretization of the continuous Markov chain

Table 3: List of initialization parameters

Parameter	variable name	role
	`init_with_I`	value of $l(c)$ under which the cell is initialized as I
R	`init_R`	initial radius of the tumor
	`init_nb_cells`	initial number of cells

6 Results

Only minimal testing and analysis has been done on the model because of time constraints, and because the model did not behave the way it was expected. Crucially, no sets of parameters have been found to produce clear chemotactic migration of the cells. The model suffers from a quite high dimensional parameter space which makes it hard to explore. The output space of the model is also very large, with very diverse behavior that are not biologically relevant (dissolution of the cells, loss of cell connectedness). Notably, the position of the cells is not something that is well-defined. For example, one could consider the center of mass of the cell, but this would not differentiate between a migrating cell and a cell dissolving in the medium and loosing all connectedness. Therefore, performing quantitative statistical analysis would have first required to implement a few well-thought way to measure the behavior of the simulations. This would require more time.

For this reason, only a qualitative exploration of the model has been done. First, let's note that the effect of the temperature parameter is redundant. Indeed, each term in the hamiltonian H is multiplied by one of the paramters. Changing T is therefore equivalent to changing all those parameters proportionally.

Each experiment was performed with at least 10^6 iterations, and some went up to 5×10^6 iterations.

6.1 Sensitivity analysis

This sensitivity uses the one-at-a-time method. Starting from the set of parameters described in the file `param_oat.txt`, each parameter was changed. The number of change and their value was not subject to any systematic method. Qualitative observed changes are described bellow.

- a_t : changes the size of all cells. Smaller cells are more sensitive to random chance, so smaller values tend to produce more disordered results, with vanishing cells and loss of connectedness.
- λ_{area} : low value tends to produce small cells that are not round and are often very disordered, with loss of connectedness. High values produce a more static system, and increase the probability to fall in a local minimum with no transition ever accepted.
- p_t : low values produce a more static system with very round cells as they try to minimize their perimeter while keeping their area. High values favorize loss of connectedness and cell with thin shapes.
- λ_{perim} : disabling the perimeter potential creates disordered cell, with loss of connectedness. High values a more static system where cells organise as tiling tilted rectangles.
- $J_{I,I}, J_{I,N}, J_{N,N}$: no clear impact, high values increase the probability that the system falls in a local minimum where no transition is ever accepted.
- D_s : directly controls the spread of the S signal. No apparent impact on the behavior of the cell.
- d_s : directly controls the decay of the S signal. No apparent impact on the behavior of the cell.
- e_s : proportionally scales the concentration function. No apparent impact on the behavior of the cell.
- $t_{N \to I}$: directly controls the amont of time before all cells of the perimeter of the tumor are of type I.
- $t_{I \to N}$: low values don't have a clear impact. High values prevent the formation of long-lasting I cells until the random motion of cells produces cells that are not connected to the tumor.
- μ_I : no clear impact
- ξ_I : no clear impact

6.2 Removing the CTMC

The reason the model failed at producing cell migration could have been the fact that all cells tend to obtain phenotype I after some time simulating. This results in no more S signal being produced, and the loss of the

concentration gradient. To test this hypothesis, the parameter set `param_no_CTMC.txt` disables all phenotype transitions and sets some cells on the perimeter of the tumor to phenotype I at the start of the simulation. This way, the tumor keeps a set of N cells that can sustain the S concentration gradient. Even with this set of parameters, no significant cell migration was observed.

7 Conclusion

It is too early to deduce any results from this work. More care should first be used to make sure the results obtained are indeed the correct outcomes of the model, and not caused by some implementation mistakes. Indeed, as the deadline for this homework was coming the code quality strongly decreased. One way to check both the model and the implementation would be to try to reproduce results obtained from models published in the litterature that take into account chemotaxis. The implementation of this model is sufficiently general that it could be modified to reproduce results from [3] with little modifications. This would definitely identify any problem with the implementation.

More work should be done to compute quantitative observables of the system trajectory, with the goal of allowing more systematic and automatic exploration of the parameter space and stop relying on the qualitative *feeling* of the experimenter. The model being inherently stochastic, more care should be employed to replicate and aggregate simulations for each set of parameters. Moreover, the length of the simulation is very important, and it might be that cell migration does happen but at an unnoticably slow rate. Finding a systematic way to simulate the system until no significant evolution could be interesting, for example by considering the evolution of H over time.

The model could be simplified. The two models surveyed in the introduction either use a potential for the area or a potential for the perimeter, but not both. It seems that it's possible to have realistic cell behavior with only one of those potential, and that the adhesion potential may be partially redundant with the area and perimeter potentials.

Finally, half of the proposed initial model was not implemented. While extending the model is not going to solve the problems encountered at this stage of the project, once those problems are fixed, the complete model might yield interesting results.

Bibliography

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